CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 4: Dynamic Programming Knapsack & Shortest Paths

Course: Design Analysis and Algorithms [23CS3502]
Category: Assignments | Assignment #4 | Academic Year: 2026-27

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1. Overview & Objective:

Construct tabular memoization solutions to eliminate redundant subproblem evaluation.

2. Assignment Problems & Instructions:

1. Formulate DP state transition for 0/1 Knapsack: V[i, w] = max V[i-1,w], v_i + V[i-1, w-w_i] .
2. Compute optimal parenthesization for Matrix Chain Multiplication of 5 dimensions.
3. Derive all-pairs shortest paths using Floyd-Warshall DP relaxation matrix.
4. Find the Longest Common Subsequence of two strings and trace back the alignment.

Submission Guidelines & Verification:

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